

Technical Document #923: Cybersecurity - Comprehensive Overview

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Penetration testing simulates cyber attacks to identify vulnerabilities. It helps organizations assess their security posture before real attacks occur.

Security information and event management (SIEM) systems collect and analyze security logs. They provide real-time visibility into security events.

Encryption converts plaintext into ciphertext to protect data confidentiality. AES and RSA are widely used encryption algorithms.

Zero trust architecture assumes no implicit trust and verifies every request. It minimizes the attack surface and limits lateral movement.

Incident response plans define how organizations respond to security incidents. They include preparation, detection, containment, and recovery phases.

Vulnerability management identifies, classifies, and remediates security vulnerabilities. Regular scanning and patching are essential components.

Federated learning trains models across decentralized data sources without sharing raw data. It enables privacy-preserving machine learning.

Natural language processing has made significant advances with large language models. These models can understand and generate human-like text with remarkable fluency.

Computer vision applications are becoming ubiquitous, from facial recognition to autonomous driving. Deep learning has enabled breakthroughs in visual understanding.

Quantum computing promises to solve problems that are intractable for classical computers. It has potential applications in cryptography, drug discovery, and optimization.

Remote work technologies have transformed the workplace. Collaboration tools and cloud services enable distributed teams to work effectively from anywhere.

Key Takeaways:

- Multi-factor authentication (MFA) requires multiple forms of verification.
- Security information and event management (SIEM) systems collect and analyze security logs.
- Intrusion detection systems (IDS) monitor network traffic for suspicious activity.